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ORIGINAL ARTICLE

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Folding behavior of the thin-walled lenticular deployable composite boom: Analytical analysis and many-objective optimization

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ABSTRACT

The thin-walled lenticular deployable composite (LDC) boom can achieve folding and deployment functions by storing and releasing strain energy. In this paper, analytical models for predicting the folding moment and the ultimate coiling radius of the lenticular DCB in the folding process were established based on the energy principle and the classical laminate theory. By using the non-dominated sorting genetic algorithm III (NSGA-III), a many-objective optimization design framework for optimizing the lenticular DCB was proposed. The optimization results show that 48 design points are found on the Pareto front, all of which are better than test sample.

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Deployable composite boom; folding behavior; analytical model; many-objective optimization; NSGA-III

Nomenclature

a	width of bonding interface, mm
A	cross-section area of the LDC boom, mm^2
b	width of completely flattened LDC boom, mm
B	transformation variable
c	projection length of the circular arc section of the cross-section, mm
E_1	longitudinal elastic modulus of the ply, GPa
E_2	transverse elastic modulus of the ply, GPa
E_θ^k	off-axis elastic modulus of the kth ply in tangential direction of θ, GPa
G_{12}	in-plane shear modulus, GPa
h	ratio of I_z to I_y
I_y	inertia moment of y axis of the cross-section, mm^4
I_z	inertia moment of z axis of the cross-section, mm^4
L	length of the LDC boom, mm
n	number of composite plies
P_c	compressive load, N
r	smaller value of $r_{1,0}$ and $r_{2,0}$
r_1	curvature radius of the concave arc, mm
$r_{1,0}$	initial curvature radius of the concave arc, mm
r_2	curvature radius of the convex arc, mm
$r_{2,0}$	initial curvature radius of the convex arc, mm
R	coiling radius, mm
R_{ultimate}	ultimate coiling radius, mm
S	linear density, g/cm
S_{12}	in-plane shear strength of composite ply, MPa
t	thickness of the LDC boom shell, mm
T	folding moment, N·m
U	total strain energy of the LDC boom in the folding process
U_0	strain energy of the LDC boom in the flattening state at one end
U_1	strain energy of the LDC boom in the completely flattening state
U_2	strain energy of the LDC boom in the coiling process
W	external work of the LDC boom in the folding process
X_t	longitudinal tensile strength of the ply, MPa
X_c	longitudinal compressive strength of the ply, MPa
Y_t	transverse tensile strength of the ply, MPa
Y_c	transverse compressive strength of the ply, MPa
φ	central angle of neutral axis on cross-section, rad
φ_0	initial central angle of neutral axis on cross-section, rad
δ	displacement at the apex of flattened half LDC boom in vertical direction, mm
θ	ply angle, deg
ν_{12}	Poisson's ratio of the ply
ν_{21}	Poisson's ratio of the ply
σ_1^k	maximum longitudinal principle stress of the kth ply in the laminate, MPa
σ_2^k	maximum transverse principle stress of the kth ply in the laminate, MPa
σ_x^k	maximum stress of the kth ply in the x direction, MPa
σ_y^k	maximum stress of the kth ply in the y direction, MPa
τ_{12}^k	maximum principle shear stress of the kth ply in the laminate, MPa
τ_{xy}^k	maximum shear stress of the kth ply in the x-y direction, MPa
awGA	adaptive-weight genetic algorithm
ES	effect size
mHV	hyper volume
mIGD	inverted generational distance
LDC	lenticular deployable composite
MOEA/D	multi-objective evolutionary algorithm based on decomposition
NASA	National Aeronautics and Space Administration
NSGA-II	non-dominated sorting genetic algorithm II
NSGA-III	non-dominated sorting genetic algorithm III
RVEA	reference vector guided evolutionary algorithm

1. Introduction

Due to the ability to overcome the strict volume limitation of vehicles, thin-walled deployable composite structures are increasingly considered for space deployable structures

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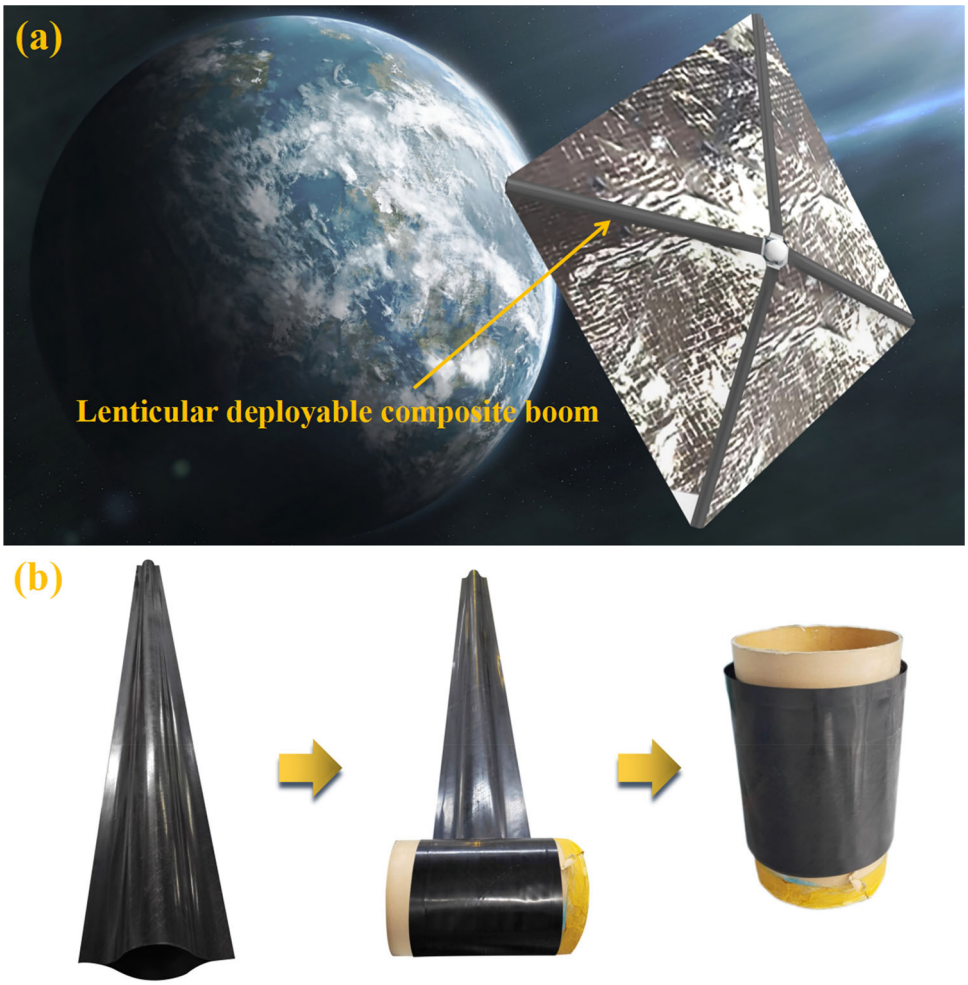


Figure 1. Typical application and folding deformation process of the LDC boom (a) Support bars of deployable solar sails (b) Folding deformation process.

[1–6], such as hinges, support bars and booms, etc. A notable example is the thin-walled lenticular deployable composite (LDC) boom, which is typically fabricated by bonding thin-walled curved composite shells. The LDC boom can not only be packed into a small space by folding, but also be restored to its initial configuration by releasing the strain energy stored in the folding process, as shown in Figure 1. The LDC boom can be employed as a basic part of space deployable structures, which has the advantages of lightweight, high stiffness and repeatability. In addition, it can also be utilized as a structural backbone component in space deployable parabolic mesh antennas, deployable solar sails and space probes, as shown in Figure 1a.

Since National Aeronautics and Space Administration (NASA) first proposed the concept of the LDC boom, the largest space agencies and corporations are studying the space application of the LDC boom. After the joint development of NASA and European Space Agency, the LDC boom was used in the Ulysses mission for the first time [7]. In February 2009, Deutsches Zentrum für Luft-und Raumfahrt developed a LDC boom and proved that the LDC boom can be used as a reliable component for long-term space missions under zero-g-conditions [8]. European Aeronautic Defense and Space Company also designed and prepared various space deployable structures based on the LDC boom [9]. In 2017, NASA designed a solar array based on the LDC boom. The solar array can release the strain energy stored in the folding process of the LDC boom to achieve its deployment function. Compared with traditional deployable solar arrays, the solar array has the advantages of high flexibility and lightweight, and it had been validated by experiments on the International Space Station [10]. Recently, NASA has also recently expressed an interest in the light and small LDC boom as a candidate solar sail boom for low-cost deep space exploration [11].

The mechanical properties of the LDC boom have gradually become a research hotspot in recent years. The folding behavior is one of the most important mechanical properties of the LDC boom. The LDC boom is subjected to various kinds of mechanical loads in the folding process, illustrating complex and nonlinear mechanical responses. Lee et al. [12, 13] established a non-tensile analytical model to predict the extensional folding deformation of the LDC boom. Based on the established analytical model, a design scheme which can realize the bistable function of the LDC boom was determined, as well as to control the folding and deployment processes. Bai et al. [14] proposed an analytical model for predicting the flattening and coiling deformation of the LDC boom. The LDC boom specimen prepared by the vacuum bag method was used as validation method and

Table 1. A summary of the existing works included in review.

Refs	Methods	Pros	Cons
[12,13]	Analytical modeling, optimization and experiment	1. Accurate modeling 2. Successful experiment	1. Simple parametric analysis 2. Without the folding moment and the ultimate coiling radius
[14]	Analytical modeling and experiment	1. Accurate modeling 2. Successful experiment	Without the folding moment and the ultimate coiling radius
[15]	Numerical modeling	1. Accurate modeling 2. Detailed description	Without the ultimate coiling radius
[16]	Optimization, analytical and numerical modeling	1. Accurate modeling 2. Comprehensive optimization process	1. Simple-objective optimization 2. Without the folding moment and the ultimate coiling radius
[17]	Numerical modeling and experiment	1. Accurate modeling 2. Successful experiment	Without the folding moment and the ultimate coiling radius

there was a good correlation between the results predicted and the experiments. In the subsequent work, numerical simulations were performed to predict extension flattening, compression flattening and coiling processes of the LDC boom [15]. Similar studies were also reported by Chu and Lei [16]. Liu et al. [17] designed and manufactured a LDC boom based on shape memory composite materials. The folding and deployment processes as well as stress distribution of the LDC boom were analyzed through experiments and numerical simulation. Table 1 summarizes the pros and cons of existing works.

The folding moment and the ultimate coiling radius are the key factors to evaluate the folding behavior of the LDC boom. The folding process of the LDC boom is a typical large deformation problem, which brings great challenges to the mechanical modeling analysis. It is clear from Table 1 that some researchers have studied the folding moment and ultimate coiling radius of the LDC boom through experimental method and numerical simulation, while the two mechanical properties predicted by analytical models have not been reported. In addition, some researchers just use simple methods to optimize the LDC boom, including simple parametric analysis and single-objective optimization design. However, the LDC boom often needs to show excellent performance in some aspects to satisfy the requirements of different space applications, such as high flexural stiffness, high natural frequency, lightweight and high packaging efficiency, etc. For these severely conflicting performance indices, many-objective optimization can output a set of flexible solutions [18–21]. Unlike the single-objective optimization problem which only provides a single optimal solution, the many-objective optimization problem will provide a Pareto front set, which represents the tradeoff solution between conflicting objectives.

Motivated by the shortcomings of related work listed in Table 1, the main goal of this paper is to establish analytical models for predicting the folding behavior of the LDC boom and perform many-objective optimization design. In this paper, based on the energy principle and the maximum stress criterion, analytical models for predicting the folding moment and the ultimate coiling radius of the LDC boom were established, which were consistent with experimental results. Analytical models are practical and convenient, and only few geometric parameters and mechanical properties of constituent materials are required to predict the folding behavior, which has a good practical value in engineering. Furthermore, a many-objective optimization design framework for the LDC boom was proposed based on non-dominated sorting genetic algorithm III (NSGA-III) algorithm. Using the proposed many-objective optimization design framework, the efficiency of searching the optimal designs of the LDC boom will be significantly increased in the practice. Finally, five state-of-the-art algorithms were benchmarked through two performance metrics and Friedman test. It is shown that the selection of the algorithm is crucial and non-dominated sorting genetic algorithm III (NSGA-II) and NSGA-III ranked the top two algorithms on the optimization design of the LDC boom. The paper is organized as follows the geometrical model of the LDC boom is established in Section 2; an analytical for predicting the folding moment of the LDC boom is proposed and it is validated using the experimental result in Section 3; an analytical model to predict the ultimate coiling radius is presented in Section 4; many-objective optimization design schemes are detailed in Section 5; the optimization results are described in Section 6; the comparison of the five state-of-the-art algorithms are discussed in Section 7; Results and discussion are shown in Section 8; four key findings are summarized in Section 9.

2. Geometrical model

The LDC boom is typically composed of two thin-walled curved composite shells, which can be folded into a roll for packaging by employing a roller, as shown in Figure 1b. The LDC boom can recover to the initial configuration by releasing the stored elastic strain energy. To characterize the geometric behavior of the LDC boom in the folding deformation, as in the literature [15], the following basic assumptions are listed below:

1. Thickness changes of the LDC boom shell is negligible small during the folding process, so the overall deformation can be described with the changes of shape and curvature of the neutral surface without local extension.
2. The LDC boom in the folding process is approximated as a thin-walled curved beam with lenticular cross-section described by the concave and convex arcs.
3. The folding deformation of the LDC boom is idealized as the linear superposition of both the quasi-static elastic flattening and coiling deformation modes, which

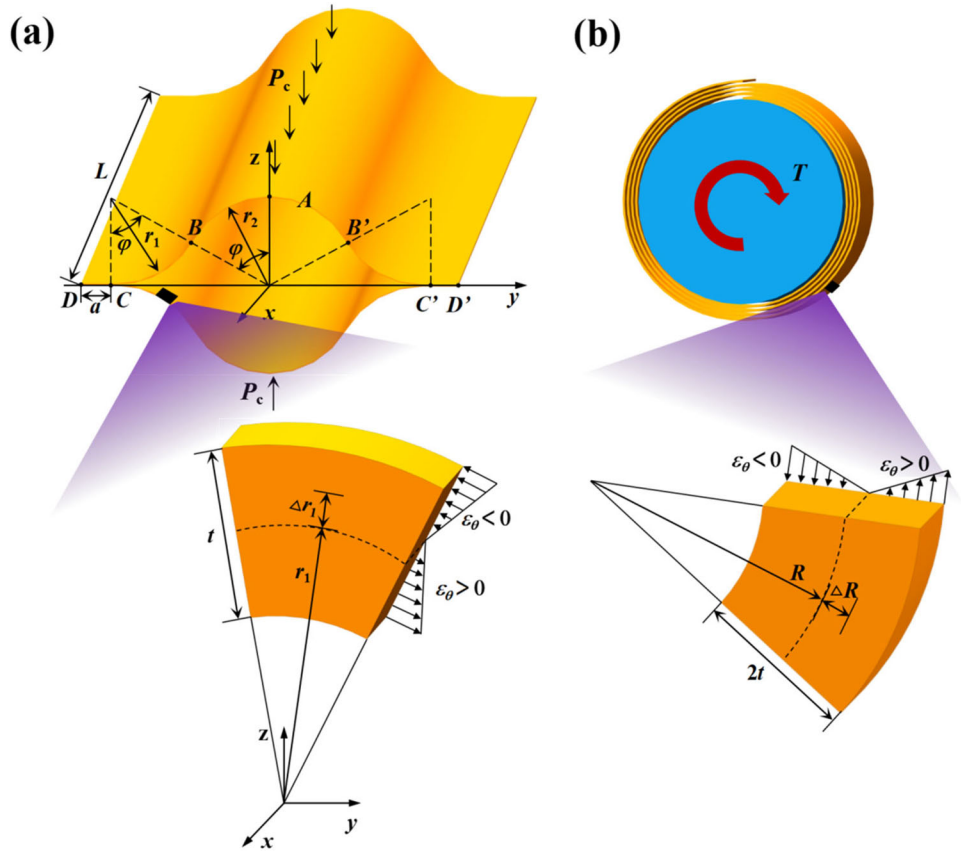


Figure 2. Geometrical configurations and representative unit cells of the LDC boom in the folding process: (a) Flattening process (b) Coiling process.

remain identical along longitudinal and transverse directions respectively.

Based on the above basic assumptions, the polar coordinate systems are chosen as shown in Figure 2a and 2b. To depict the geometrical configuration of the LDC boom in the flattening and coiling processes, the representative unit cells are selected, as shown in Figure 2a and 2b. From Figure 2, the flattening deformation of the LDC boom can be described by the central angle of neutral axis on cross-section in the polar coordinate system, and the coiling deformation of the LDC boom can be depicted by the curvature of the completely flattening LDC boom in the x direction. Therefore, the following geometrical model for describing the folding deformation of the LDC boom can be obtained.

According to assumptions (1) and (2), the geometrical constraint to be satisfied is

$$\cos\varphi = \frac{r_2}{r_1 + r_2} \tag{1}$$

where r_1 and r_2 are the curvature radii of the concave and convex arcs respectively, and φ is the central angle of neutral axis on cross-section.

The half LDC boom is composed of two concave arcs, one convex arc and two flat adhesive edges. Therefore, the width of the completely flattening LDC boom is

$$b = 2\varphi(r_1 + r_2) + 2a \tag{2}$$

where a is the width of bonding interface.

The cross-section area of the completely flattening LDC boom can be expressed as

$$A = 2bt = 4t\varphi(r_1 + r_2) + 4ta \tag{3}$$

where t is the thickness of the LDC boom shell.

The inertia moments of the LDC boom in initial state (i.e. completely deployed state) can be written as

$$I_y = \int z^2 dA \tag{4}$$

$$I_z = \int y^2 dA \tag{5}$$

Substituting geometric parameters into Eqs. (4) and (5) and applying coordinate conversions, it is possible to show as

$$I_y = 4\int_0^{\varphi_0} (r_{2,0}\cos\beta)^2 d\beta r_{2,0} t + 4\int_0^{\varphi_0} \left[r_{1,0}(1-\cos\alpha)\right]^2 d\alpha r_{1,0} t + \frac{at^3}{3} \tag{6}$$

$$I_z = 4\int_0^{\varphi_0} (r_{2,0}\sin\beta)^2 d\beta r_{2,0} t + 4\int_0^{\varphi_0} (r_{1,0}\sin\varphi_0 + r_{2,0}\sin\varphi_0 - r_{1,0}\sin\alpha)^2 d\alpha r_{1,0} t + \frac{ta^3}{3} \tag{7}$$

where $r_{1,0}$ and $r_{2,0}$ are the initial curvature radii of the concave and convex arcs respectively, and $\varphi\sigma\upsilon\beta > 0\ \backslash\sigma\upsilon\beta >$ is the initial central angle of neutral axis on cross-section.

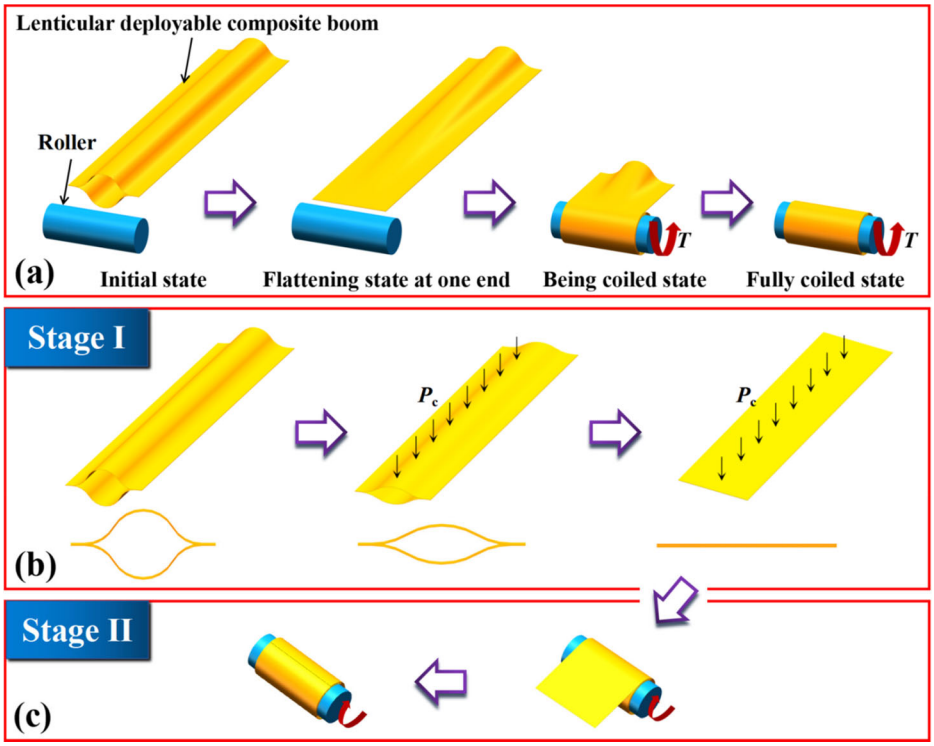


Figure 3. Schematic diagram of folding process of the LDC boom: (a) Practical folding process (b) Flatting process (b) Coiling process.

Integrating Equations (6) and (7) they take the form below:

$$I_y = r_{2,0}^3 t(2\varphi_0 + \sin 2\varphi_0) + r_{1,0}^3 t(6\varphi_0 - 8\sin\varphi_0 + \sin 2\varphi_0) + \frac{at^3}{3} \tag{8}$$

$$I_z = \left(r_{1,0}^3 t + r_{2,0}^3 t\right)(2\varphi_0 - \sin 2\varphi_0) + 4r_{1,0} t\varphi_0 \sin^2\varphi_0 (r_{1,0} + r_{2,0})^2 + 8r_{1,0}^2 t \sin\varphi_0(\cos\varphi_0 - 1)(r_{1,0} + r_{2,0}) + \frac{ta^3}{3} \tag{9}$$

3. Analytical model for folding moment

3.1. Analytical model

The schematic diagram of folding process of the LDC boom is shown in Figure 3a to 3c. The folding process is divided into two stages, as shown in Figure 3b and 3c. In the first stage, a compressive load is applied at the top and bottom ends of the LDC boom. In this stage, the external work is converted into strain energy, as shown in Figure 3b; in the second stage, a torque is employed on the roller to coil the LDC boom, the curvature in the length direction changes from 0 to $1/R$, as shown in Figure 3c.

In the first stage, the compressive load-displacement curve of the half LDC boom can be expressed by the parametric equation of φ, r_1 and r_2 [14].

$$\begin{cases} \delta(\varphi, r_1, r_2) = (r_{1,0} + r_{2,0})(1 - \cos\varphi_0) - (r_1 + r_2)(1 - \cos\varphi) \\ P_c(\varphi, r_1, r_2) = \dfrac{4r_{1,0}BL(\varphi - \varphi_0)}{(r_1 + r_2)\varphi_0 \sin\varphi} \end{cases}, \left(\varphi \in [\varphi_0, 0], r_1 \in [r_{1,0}, +\infty], r_2 \in [r_{2,0}, +\infty]\right) \tag{10}$$

where L is the length of the LDC boom and B is the transformation variable.

$$B = \sum_{k=1}^{n} \left\{ E_\theta^k \left[\left(\Delta r_1^k - \Delta r_1^{k-1}\right) - r_{1,0} \ln \frac{r_{1,0} + \Delta r_1^k}{r_{1,0} + \Delta r_1^{k-1}} \right] \right\} \tag{11}$$

where n is the number of composite plies and θ is the ply angle. E_θ^k is the off-axis elastic modulus of the kth ply in tangential direction of θ, which can be written as [22]

$$\frac{1}{E_\theta^k} = \frac{1}{E_1}\sin^4\theta^k + \left(\frac{1}{G_{12}} - \frac{2\nu_{12}}{E_1}\right)\sin^4\theta^k \cos^4\theta^k + \frac{1}{E_2}\cos^4\theta^k \tag{12}$$

According to assumption (1), although there is a large deformation in the flattening process and the geometry as well as curvature of the LDC boom have changed significantly, the total length of the neutral surface has not changed. Thus, it is assumed to have

$$\varphi(r_1 + r_2) = \varphi_0(r_{1,0} + r_{2,0}) \tag{13}$$

Substituting Eq. (13) into Eq. (10) and eliminating r_1+r_2, the parametric equation of φ can be obtained as

$$\begin{cases} \delta(\varphi) = (r_{1,0} + r_{2,0})(1 - \cos\varphi_0) - \dfrac{\varphi_0}{\varphi}(r_{1,0} + r_{2,0})(1 - \cos\varphi) \\ P_c(\varphi) = \dfrac{4r_{1,0}\varphi BL(\varphi - \varphi_0)}{(r_{1,0} + r_{2,0})\varphi_0^2 \sin\varphi} \end{cases}, (\varphi \in [\varphi_0, 0]) \tag{14}$$

The external work of the LDC boom when compressed from the initial state to the completely flattening state is twice the definite integral of parametric Eq. (14), that is:

$$U_1 = 2\int_{\varphi_0}^{0} P_c(\varphi)\delta'(\varphi)d\varphi \tag{15}$$

According to the energy principle, the strain energy of the LDC boom in the completely flattening state is equal to the external work. Substituting Eq. (14) into Eq. (15), the strain energy of the LDC boom in the completely flattening state can be obtained as

$$U_1 = \int_{\varphi_0}^{0} \frac{8r_{1,0}BL(\varphi_0 - \varphi)(\varphi\sin\varphi + \cos\varphi - 1)}{\varphi\varphi_0 \sin\varphi} d\varphi \tag{16}$$

Utilizing the classical laminate theory, the elastic deformation of the LDC boom in the coiling process can be expressed as

$$\begin{bmatrix} \mathbf{N} \\ \mathbf{M} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \boldsymbol{\varepsilon_0} \\ \boldsymbol{\kappa} \end{bmatrix} \tag{17}$$

The completely flattening LDC boom is only subjected to the bending moment in the x direction in the coiling process, so it is possible to determine as

$$\begin{cases} \mathbf{N} = \mathbf{0} \\ M_y = 0 \\ M_{xy} = 0 \\ \kappa_y = 0 \\ \kappa_{xy} = 0 \end{cases} \tag{18}$$

Standard composite construction makes use almost exclusively of symmetric laminates, which have both geometric and material symmetry about the middle surface, i.e. $\boldsymbol{B} = 0$.

Substituting Eq. (18) into Eq. (17) can be obtained as

$$\begin{bmatrix} M_x \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} \kappa_x \\ 0 \\ 0 \end{bmatrix} \tag{19}$$

Since the LDC boom is in the pure bending state during the coiling process, the strain energy of the LDC boom in the coiling process is

$$U_2 = \frac{bL}{2} \begin{bmatrix} \kappa_x & 0 & 0 \end{bmatrix} \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} \kappa_x \\ 0 \\ 0 \end{bmatrix} \tag{20}$$

In the coiling process, the curvature in the x direction of the LDC boom can be written as

$$\kappa_x = \frac{1}{R} \tag{21}$$

where R is the coiling radius and $\boldsymbol{D}$ is the stiffness matrix.

Substituting Eqs. (2), (13) and (21) into Eq. (20) and after simplification, it is possible to have

$$U_2 = \frac{D_{11}L(\varphi_0 r_{1,0} + \varphi_0 r_{2,0} + a)}{R^2} \tag{22}$$

In the practical folding process of the LDC boom, one end is flattened in the first step, and then the folding process is completed through a roller in the second step. The schematic diagram of the whole folding deformation process is shown in

Figure 3a. In the first step of the compressive deformation process of the LDC boom, except the cross-section shape of the neutral cross-section, the cross-section shapes of the other positions along the length direction change. The cross-section from the neutral cross-section to the compressive end will be flattened in the width direction, and the cross-section from the neutral cross-section to the free end will be shortened. The position of the neutral cross-section of the LDC boom in the compressive deformation process is at 2/3 of the length direction, and the projection length of the circular arc section of the cross-section is linearly distributed along the length direction [23].

At the compressive end ($x=0$), the projection length of the circular arc section of the cross-section is

$$c|_{x=0} = 2(r_{1,0} + r_{2,0})\varphi_0 \tag{23}$$

At the neutral cross-section ($x = \frac{2}{3}L$), the projection length of the circular arc section of the cross-section is

$$c|_{x=\frac{2}{3}L} = 2(r_{1,0} + r_{2,0}) \sin \varphi_0 \tag{24}$$

Therefore, the relationship between $c(x)$ and x can be deduced as

$$c(x) = \frac{3}{L}(r_{1,0} + r_{2,0})(\sin \varphi_0 - \varphi_0)x + 2(r_{1,0} + r_{2,0})\varphi_0, (x \in [0, L]) \tag{25}$$

According to the geometric configuration, it is possible to have

$$c(x) = 2[r_{1,1}(x) + r_{2,1}(x)] \sin \varphi_1(x), (x \in [0, L]) \tag{26}$$

$$\varphi_1(x)[r_{1,1}(x) + r_{2,1}(x)] = \varphi_0(r_{1,0} + r_{2,0}), (x \in [0, L]) \tag{27}$$

Substituting Eqs. (26) and (27) into Eq. (25) and eliminating $r_{1,1}(x)+r_{2,1}(x)$, the relationship between x and $\varphi_1(x)$ can be expressed as

$$\frac{\sin \varphi_1(x)}{\varphi_1(x)} = \frac{3(\sin \varphi_0 - \varphi_0)}{2\varphi_0 L}x + 1, (x \in [0, L]) \tag{28}$$

Obviously, Eq. (28) is an implicit function, which can be solved numerically by Newton iterative method.

According to Eq. (16) and the definition of double integral, the strain energy of the LDC boom in the flattening state at one end can be written as

$$U_0 = \int_0^L \left| \int_{\varphi_0}^{\varphi_1(x)} \frac{8r_{1,0}B(\varphi_0 - \varphi)(\varphi \sin \varphi + \cos \varphi - 1)}{\varphi\varphi_0 \sin \varphi} d\varphi \right| dx \tag{29}$$

The total strain energy of the LDC boom in the folding process can be expressed as

$$U = U_1 + U_2 - U_0 \tag{30}$$

Substituting Eqs. (16), (22) and (29) into Eq. (30), it is possible to determine as

$$\begin{aligned} U = &\int_{\varphi_0}^{0} \frac{8r_{1,0}BL(\varphi_0 - \varphi)(\varphi \sin \varphi + \cos \varphi - 1)}{\varphi\varphi_0 \sin \varphi} d\varphi + \frac{D_{11}L(\varphi_0 r_{1,0} + \varphi_0 r_{2,0} + a)}{R^2} \\ &- \int_0^L \left| \int_{\varphi_0}^{\varphi_1(x)} \frac{8r_{1,0}B(\varphi_0 - \varphi)(\varphi \sin \varphi + \cos \varphi - 1)}{\varphi\varphi_0 \sin \varphi} d\varphi \right| dx \end{aligned} \tag{31}$$

The external work of the LDC boom in the folding process can be written as

$$W = \frac{TL}{R} \tag{32}$$

where T is the folding moment of the LDC boom in the folding process.

According to the energy principle, the total strain energy of the LDC boom in the folding process is equal to the external work. Thus, the folding moment of the LDC boom in the folding process can be obtained by substituting Eq. (31) in Eq. (32), as follows

$$\begin{aligned} T = &\int_{\varphi_0}^{0} \frac{8r_{1,0}BR(\varphi_0 - \varphi)(\varphi \sin \varphi + \cos \varphi - 1)}{\varphi\varphi_0 \sin \varphi} d\varphi + \frac{D_{11}(\varphi_0 r_{1,0} + \varphi_0 r_{2,0} + a)}{R} \\ &- \frac{R}{L}\int_0^L \left| \int_{\varphi_0}^{\varphi_1(x)} \frac{8r_{1,0}B(\varphi_0 - \varphi)(\varphi \sin \varphi + \cos \varphi - 1)}{\varphi\varphi_0 \sin \varphi} d\varphi \right| dx \end{aligned} \tag{33}$$

Table 2. Specifications and properties of carbon/epoxy composite ply [15].

Specifications and properties	Values
Longitudinal elastic modulus E_1 (GPa)	80.08
Transverse elastic modulus E_2 (GPa)	6.67
In-plane shear modulus G_{12} (GPa)	2.93
Inter-laminar shear modulus G_{13} (GPa)	2.93
Inter-laminar shear modulus G_{23} (GPa)	2.50
Poisson's ratio $\nu\sigma\upsilon\beta > 12/\sigma\upsilon\beta >$	0.344
Poisson's ratio $\nu\sigma\upsilon\beta > 21/\sigma\upsilon\beta >$	0.036
Longitudinal tensile strength X_t (MPa)	1063.48
Transverse tensile strength Y_t (MPa)	65.99
Longitudinal compressive strength X_c (MPa)	662.69
Transverse compressive strength Y_c (MPa)	144.54
In-plane shear strength S_{12} (MPa)	126.76
Density ρ (g/mm^3)	1.60×10^{-3}
Ply thickness (mm)	0.08

Table 3. Specifications of the LDC boom [15].

Geometric parameters	Values
Initial curvature radius of the concave arc $r_{1,0}$ (mm)	34
Initial curvature radius of the convex arc $r_{2,0}$ (mm)	34
Width of bonding interface a (mm)	10
Length of the LDC boom L (mm)	1020
Thickness of the LDC boom shell t (mm)	0.4
Coiling radius R (mm)	80
Number of composite plies n	5
Ply angle θ (deg)	[45/-45/0/-45/45]

3.2. Model validation

In order to validate the effectiveness of the analytical model proposed in Section 3.1, the results are compared to the experimental results from the literature [15]. Specifications and mechanical properties of the carbon/epoxy composite ply are listed in Table 2, and geometric parameters of the LDC boom are listed in Table 3. Figure 4 illustrates the folding moment versus rotational displacement curves of the LDC boom in the folding process. According to Figure 4, the folding moment measured by the experiment increases with increase of the rotational displacement within the initial small deformation range and then reaches a stable level. This means that the folding deformation proceeds stably. It is clear from Figure 4 that the folding moment versus rotational displacement curve predicted by the analytical model is in good agreement with the experimental result, in which the folding moment measured in the experiment is about 1 N·m and the folding moment predicted by the analytical model is 1.289 N·m. In conclusion, the analytical model proposed in Section 3.1 accurately predicts the folding moment of the LDC boom.

4. Analytical model for ultimate coiling radius

As the LDC boom is gradually being flattened, the curvatures of the concave and convex arcs change from $1/r_{1,0}$ and $1/r_{2,0}$ to 0. In the flattening process, the arc with greater curvature variation produces greater strain, and then produces greater stress. Therefore, the curvature variation of the arc with smaller curvature radius is

$$\kappa_y = \frac{1}{r} \tag{34}$$

where

$$r = \min\{r_{1,0}, r_{2,0}\} \tag{35}$$

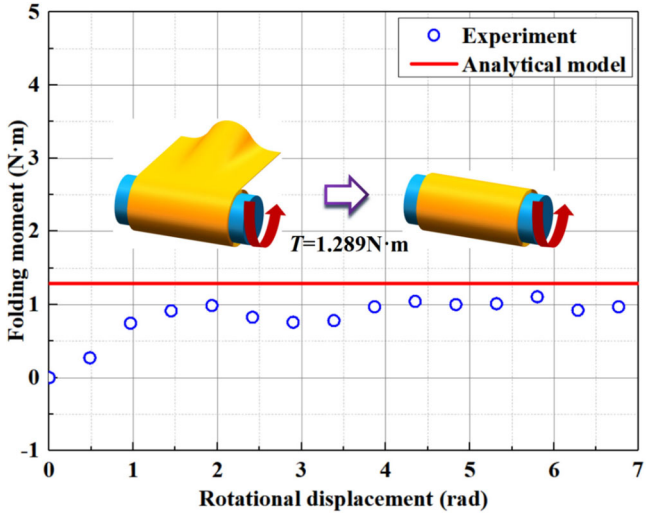


Figure 4. Folding moment versus rotational displacement curves of the LDC boom in the folding process.

When the LDC boom is coiled with the ultimate coiling radius, the curvature variation in the x direction can be expressed as

$$\kappa_x = \frac{1}{R_{ultimate}} \tag{36}$$

where R_{ultimate} is the ultimate coiling radius of the LDC boom.

The stress-strain relationship of the kth ply ($k = 1,2,3 \ldots n$) in the laminate is

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k = \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix}_k \begin{bmatrix} \kappa_x \Delta R_{ultimate} \\ \kappa_y \Delta r \\ 0 \end{bmatrix} \tag{37}$$

Substituting Eqs. (34) and (36) into Eq. (37), it is deduced as

$$\sigma_x^k = \frac{\Delta R_{ultimate} \bar{Q}_{11}^k}{R_{ultimate}} + \frac{\Delta r \bar{Q}_{12}^k}{r} \tag{38}$$

$$\sigma_y^k = \frac{\Delta R_{ultimate} \bar{Q}_{12}^k}{R_{ultimate}} + \frac{\Delta r \bar{Q}_{22}^k}{r} \tag{39}$$

$$\tau_{xy}^k = \frac{\Delta R_{ultimate} \bar{Q}_{16}^k}{R_{ultimate}} + \frac{\Delta r \bar{Q}_{26}^k}{r} \tag{40}$$

According to the coordinate transformation equation of stress components of the kth ply in the laminate, the maximum principle stresses of the kth ply in the laminate can be expressed as

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix}_k = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & -2\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}_k \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}_k \tag{41}$$

Substituting Eqs. (38) to (40) into Eq. (41), it is possible to determine as

$$\sigma_1^k = \frac{\Delta R_{ultimate}}{R_{ultimate}} \left(\cos^2\theta^k \bar{Q}_{11}^k + \sin^2\theta^k \bar{Q}_{12}^k + 2\sin\theta^k \cos\theta^k \bar{Q}_{16}^k \right) + \frac{\Delta r}{r} \left(\cos^2\theta^k \bar{Q}_{12}^k + \sin^2\theta^k \bar{Q}_{22}^k + 2\sin\theta^k \cos\theta^k \bar{Q}_{26}^k \right) \tag{42}$$

$$\sigma_2^k = \frac{\Delta R_{ultimate}}{R_{ultimate}}\left(\sin^2\theta^k\bar{Q}_{11}^k + \cos^2\theta^k\bar{Q}_{12}^k - 2\sin\theta^k\cos\theta^k\bar{Q}_{16}^k\right) + \frac{\Delta r}{r}\left(\sin^2\theta^k\bar{Q}_{12}^k + \cos^2\theta^k\bar{Q}_{22}^k - 2\sin\theta^k\cos\theta^k\bar{Q}_{26}^k\right) \tag{43}$$

$$\tau_{12}^k = \frac{\Delta R_{ultimate}}{R_{ultimate}}\left[-\sin\theta^k\cos\theta^k\bar{Q}_{11}^k + \sin\theta^k\cos\theta^k\bar{Q}_{12}^k + (\cos^2\theta^k - \sin^2\theta^k)\bar{Q}_{16}^k\right] + \frac{\Delta r}{r}\left[-\sin\theta^k\cos\theta^k\bar{Q}_{12}^k + \sin\theta^k\cos\theta^k\bar{Q}_{22}^k + (\cos^2\theta^k - \sin^2\theta^k)\bar{Q}_{26}^k\right] \tag{44}$$

The maximum stress criterion is utilized to predict the ultimate coiling radius of the LDC boom in the folding process. The critical stress state of the kth ply with tensile failure in the longitudinal (fiber) direction is

$$\sigma_1^k = X_t \tag{45}$$

Substituting Eq. (42) into Eq. (45), the ultimate coiling radius corresponding to the tensile failure of the kth ply in the longitudinal direction can be obtained, as follows

$$R_{ultimate}^{1,k} = \frac{r\Delta R_{ultimate}\left(\cos^2\theta^k\bar{Q}_{11}^k + \sin^2\theta^k\bar{Q}_{12}^k + 2\sin\theta^k\cos\theta^k\bar{Q}_{16}^k\right)}{X_t r - \Delta r\left(\cos^2\theta^k\bar{Q}_{12}^k + \sin^2\theta^k\bar{Q}_{22}^k + 2\sin\theta^k\cos\theta^k\bar{Q}_{26}^k\right)} \tag{46}$$

The critical stress state of the kth ply with compressive failure in the longitudinal (fiber) direction is

$$\sigma_1^k = X_c \tag{47}$$

Substituting Eq. (42) into Eq. (47), the ultimate coiling radius corresponding to the compressive failure of the kth ply in the longitudinal direction can be obtained, as follows:

$$R_{ultimate}^{2,k} = \frac{r\Delta R_{ultimate}\left(\cos^2\theta^k\bar{Q}_{11}^k + \sin^2\theta^k\bar{Q}_{12}^k + 2\sin\theta^k\cos\theta^k\bar{Q}_{16}^k\right)}{X_c r - \Delta r\left(\cos^2\theta^k\bar{Q}_{12}^k + \sin^2\theta^k\bar{Q}_{22}^k + 2\sin\theta^k\cos\theta^k\bar{Q}_{26}^k\right)} \tag{48}$$

The critical stress state of the kth ply with tensile failure in the transverse (to the fiber) direction is

$$\sigma_2^k = Y_t \tag{49}$$

Substituting Eq. (43) into Eq. (49), the ultimate coiling radius corresponding to the tensile failure of the kth ply in the transverse (to the fiber) direction can be obtained, as follows

$$R_{ultimate}^{3,k} = \frac{r\Delta R_{ultimate}\left(\sin^2\theta^k\bar{Q}_{11}^k + \cos^2\theta^k\bar{Q}_{12}^k - 2\sin\theta^k\cos\theta^k\bar{Q}_{16}^k\right)}{Y_t r - \Delta r\left(\sin^2\theta^k\bar{Q}_{12}^k + \cos^2\theta^k\bar{Q}_{22}^k - 2\sin\theta^k\cos\theta^k\bar{Q}_{26}^k\right)} \tag{50}$$

The critical stress state of the kth ply with compressive failure in the transverse (to the fiber) direction is

$$\sigma_2^k = Y_c \tag{51}$$

Substituting Eq. (43) into Eq. (51), the ultimate coiling radius corresponding to the compressive failure of the kth ply in the transverse (to the fiber) direction can be obtained, as follows:

$$R_{ultimate}^{4,k} = \frac{r\Delta R_{ultimate}\left(\sin^2\theta^k\bar{Q}_{11}^k + \cos^2\theta^k\bar{Q}_{12}^k - 2\sin\theta^k\cos\theta^k\bar{Q}_{16}^k\right)}{Y_c r - \Delta r\left(\sin^2\theta^k\bar{Q}_{12}^k + \cos^2\theta^k\bar{Q}_{22}^k - 2\sin\theta^k\cos\theta^k\bar{Q}_{26}^k\right)} \tag{52}$$

The critical stress state of the kth ply with shear failure is

$$t_{12}^k = S_{12} \tag{53}$$

Substituting Eq. (44) into Eq. (53), the ultimate coiling radius corresponding to the shear failure of the kth ply can be obtained, as follows

$$R_{ultimate}^{5,k} = \frac{r\Delta R_{ultimate}\left[-\sin\theta^k\cos\theta^k\bar{Q}_{11}^k + \sin\theta^k\cos\theta^k\bar{Q}_{12}^k + (\cos^2\theta^k - \sin^2\theta^k)\bar{Q}_{16}^k\right]}{S_{12} r - \Delta r\left[-\sin\theta^k\cos\theta^k\bar{Q}_{12}^k + \sin\theta^k\cos\theta^k\bar{Q}_{22}^k + (\cos^2\theta^k - \sin^2\theta^k)\bar{Q}_{26}^k\right]} \tag{54}$$

where X_t and X_c are longitudinal tensile and compressive strength of composite ply, Y_t and Y_c are transverse tensile and compressive strength of composite ply, and S_{12} is the in-plane shear strength of composite ply.

The maximum value of the ultimate coiling radius corresponding to five failure modes and n plies is determined as the ultimate coiling radius of the LDC boom.

$$R_{ultimate} = max\ R^{j,k}_{ultimate},\ \begin{cases} j = 1,2,3,4,5 \\ k = 1,2,3...n \end{cases} \tag{55}$$

Similarly, the ultimate coiling radius of the LDC boom is predicted using specifications and mechanical properties of the carbon/epoxy composite ply in Table 2 and geometric parameters of the LDC boom in Table 3. In the literature, the coiling radius of the LDC boom is 80 mm, which can achieve multiple folding and deploying functions without failure. The predicted ultimate coiling radius using the analytical model is 37.44 mm, less than the practical coiling radius 80 mm, so the LDC boom can be folded without any damage. This result is consistent with the experiment. In conclusion, the analytical model for predicting the ultimate coiling radius of the LDC boom is effective and feasible.

5. Optimal design schemes

5.1. Problem description

Optimal design of the LDC boom is a typical many-objective optimization problem. Unlike the single-objective optimization problem which only provides a single optimal solution, many-objective optimization is applied to identify a set of tradeoff solutions to achieve the global optimization for objective functions. The set of tradeoff solutions in the space of design variables referred to as the Pareto non-inferior solution set. The map of Pareto non-inferior solution set in the objective space forms the Pareto front.

The optimization problem is formulated as a problem of minimization. The general many-objective optimization formula can be described as

$$\begin{aligned} &Minimize\ \phi(\boldsymbol{X}) = [\phi_1(\boldsymbol{X}), \phi_2(\boldsymbol{X})...\phi_i(\boldsymbol{X})] \\ &subject\ to: \\ &\quad \begin{cases} \boldsymbol{X} = (x_1, x_2...x_m)^T \\ x_g \le x_g \le \overline{x_g} (1 \le g \le m) \end{cases} \end{aligned} \tag{56}$$

where ϕ represents the objective function, $\boldsymbol{X}=(x_1,x_2 \ldots x_j)^{\mathrm{T}}$ are the design variables, i and m are respectively the number of objective functions and design variables, and x_g and $\overline{x_g}$ are the upper and lower limits of the variable space, respectively.

In practical engineering applications of LDC boom, researchers focus on the folding behavior, lightweight, flexural stiffness and natural frequency, etc. To facilitate the design and preparation of the folding device, the ultimate coiling radius should be minimized as much as possible. The folding moment should be as small as possible for folding the LDC boom more conveniently and reliably. The linear density of the LDC boom should be minimized as possible to weight and launch cost. In addition, the flexural stiffness and natural frequency mainly depend on the inertia moments of the cross-section, so the LDC boom need to have enough inertia moment to obtain high flexural stiffness and natural frequency. Therefore, a total of five objectives are set in this research, as follows

- Objective 1: minimizing the ultimate coiling radius, that is

$$\phi_1(\boldsymbol{X}) = R_{ultimate} \tag{57}$$

- Objective 2: minimizing the folding moment, that is

$$\phi_2(\boldsymbol{X}) = T \tag{58}$$

- Objective 3: minimizing the linear density, that is

$$\phi_3(\boldsymbol{X}) = \rho A \tag{59}$$

- Objective 4: maximizing the inertia moment of y axis of the cross-section, that is

$$\phi_4(\boldsymbol{X}) = -I_y \tag{60}$$

- Objective 5: maximizing the inertia moment of z axis of the cross-section, that is

$$\phi_5(\boldsymbol{X}) = -I_z \tag{61}$$

It is noting that maximizing a $\phi(X)$ function is equal to minimizing $(-\phi(X))$.

It has been illustrated in Sections 2, 3 and 4 that geometric parameters of the LDC boom and mechanical properties of constituent materials have significant effects on the ultimate coiling radius, folding moment, linear density and inertia moments of the cross-section. Compared to mechanical properties of constituent materials, geometric parameters are more suitable for optimizing the LDC boom in practical engineering applications. Therefore, three geometric parameters, namely initial curvature radii of the concave and convex arcs as well as the width of bonding interface, are taken into consideration as design variables.

$$\boldsymbol{X} = (r_{1,0}\ \ r_{2,0}\ \ a)^T \tag{62}$$

The optimization design of the LDC boom needs to satisfy some constraints. To obtain better stability, the inertia moments of y axis and z axis of the cross-section should be as close as possible. The ultimate coiling radius must be greater than the practical coiling radius to avoid failure. In addition, the linear density of the LDC boom should not be too large to avoid increasing weight and launch cost. Therefore, the constraints are described as

- Constraint 1: the geometric constraint of the inertia moment of the cross-section is

$$_h \le h = \frac{I_z}{I_y} \le \bar{h} \tag{63}$$

- Constraint 2: the LDC boom does not fail in the folding process, that is

$$R_{ultimate} \le R \tag{64}$$

- Constraint 3: the constraint of the linear density of the LDC boom is

$$S = \rho A \le \bar{S} \tag{65}$$

Penalty methods are general approaches to solve constraints in optimization problems. However, penalty methods require user-defined problem-dependent parameters, which usually have a negative effect on the performance of

the algorithm. Therefore, adaptive penalty techniques [24] have been attracted a great attention. An adaptive penalty technique for dealing with the constraints in optimization problems was employed in this article. A penalty coefficient was set in the fitness function. For any feasible solution, the penalty coefficient was 1; for any unfeasible solution, the penalty coefficient was greater than 1. Moreover, the penalty coefficient was increased with the increased magnitude of the constraint violation.

Table 4. NSGA-III algorithm parameter definition.

Initialization	**Initialization type**	**Random**
	Encoding type	**Real value coded**
Reproduction	Number of populations	200, 600, 1000, 1500
	Number of generations	50, 100, 150, 200
	Selection type	Well-spread reference points
	Crossover type	Real variable SBX
	Crossover rate	0.7
	Mutation type	Polynomial mutation
	Mutation rate	0.08
	Elitism operator	Active
Termination	Stop criterion	Reach the maximum generation

5.2. Nsga-III algorithm

NSGA-III is recognized as one of the most state-of-the-art algorithms for solving the many-objective optimization problem, which has been widely applied in various engineering problems. In this paper, NSGA-III is utilized to solve the many-objective optimization problem of the LDC boom introduced in Section 5.1. NSGA-III algorithm parameter definition is listed in Table 4, and the parameter definition is consistent with the literature [4].

5.3. Many-objective optimization design framework

Figure 5 illustrates the many-objective optimization design framework of the LDC boom with NSGA-III. The flow chart of NSGA-III algorithm based on the reference point is shown in Figure 5a. Three design variables (i.e. initial curvature radii of the concave and convex arcs as well as the width of bonding interface) and several hyper parameters are shown in Figure 5b. Five objectives are shown in Figure 5c, including the ultimate coiling radius, folding moment,

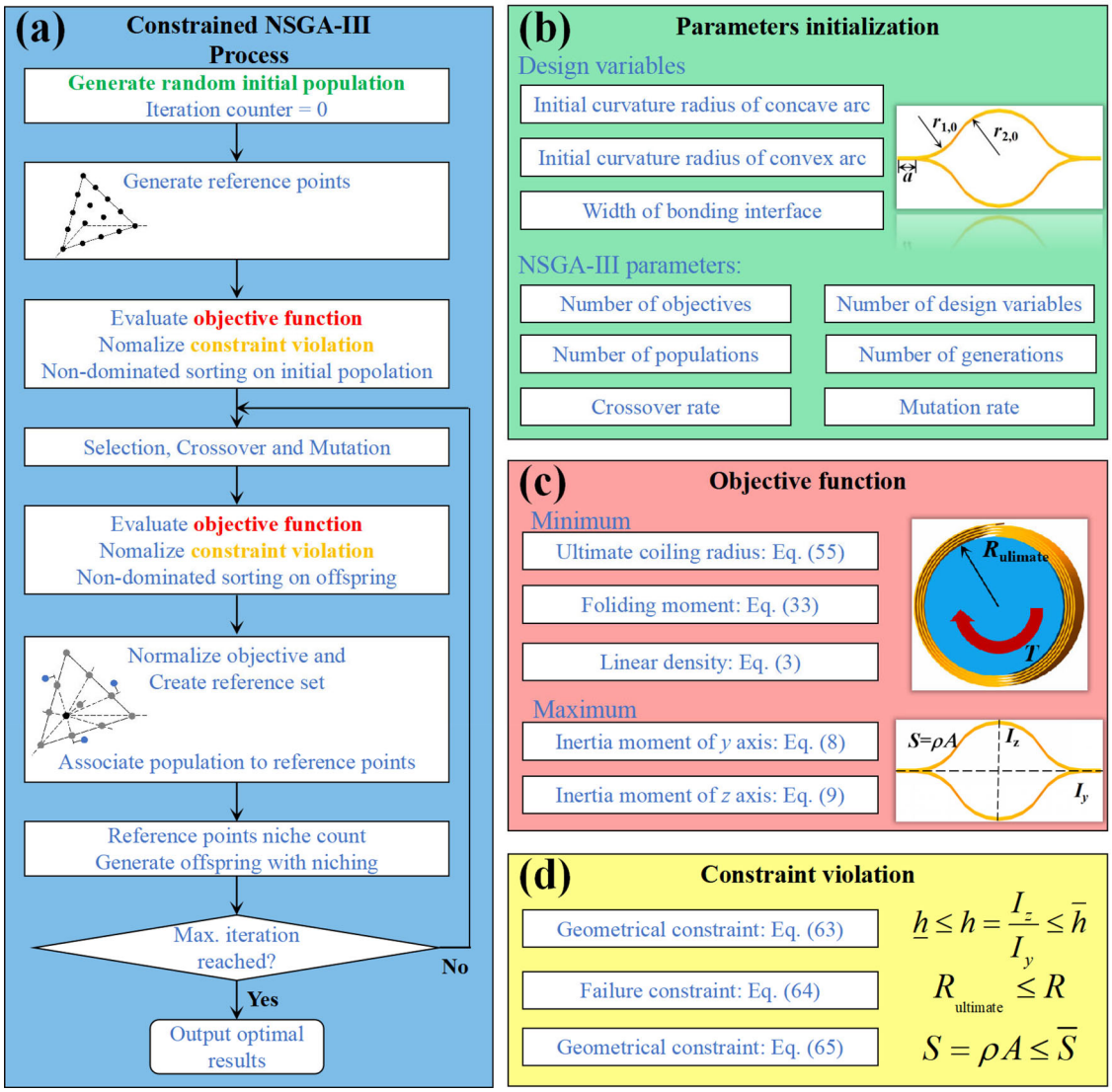


Figure 5. Many-objective optimization design framework of the LDC boom with NSGA-III: (a) Constrained NSGA-III process (b) Parameters initialization (c) Objective function (d) Constraint violation.

linear density and inertia moments of the cross-section. Three constraints are shown in Figure 5c, namely geometric constraint, no failure constraint, linear density constraint. The optimization design was achieved through a high-performance Python coded algorithm toolbox (Geatpy), which includes many state-of-the-art algorithms. Specifications and mechanical properties of the carbon/epoxy composite ply in Table 2 were employed. Moreover, except for three design variables, specifications of the LDC boom were consistent with Table 3. In order to realize a comprehensive Pareto front, the design variables were set as continuous variables. Table 5 shows the parameter setting of design variables and constraints.

Table 5. Parameter setting of design variables and constraints.

Parameter		Lower bound	Upper bound
Design variables	$r_{1,0}$(mm)	20	50
	$r_{2,0}$(mm)	20	50
	a (mm)	5	15
Constraints	h	1	5
	$R_{ultimate}$ (mm)	N/A	80
	S (g/cm)	N/A	2.50

6. Optimization results of deployable composite boom

Using the the many-objective optimization design framework of the LDC boom with NSGA-III established in Section 5, population sizes of 200, 600, 1000 and 1500 were compared to determine the optimal population size. Figure 6a–6d show the optimization results of the Pareto front. Clearly, population sizes can significantly affect the optimization results. As the population size increases, more design points appear on the Pareto front. Therefore, the results of 1500 population were selected to discuss the validity of the many-objective optimization design, as shown in Figure 7a–7c. There are a total of 1365 design points on the entire Pareto front, and the 1365 design points are sorted in ascending order of the linear density. Table 6 summarizes 22 design points to exhibit how changes in design variables affect objectives, in which the 22 designs were picked from the first point of each 65 points interval. These 22 design points illustrate that the relationship between design variables and objectives is nonlinear.

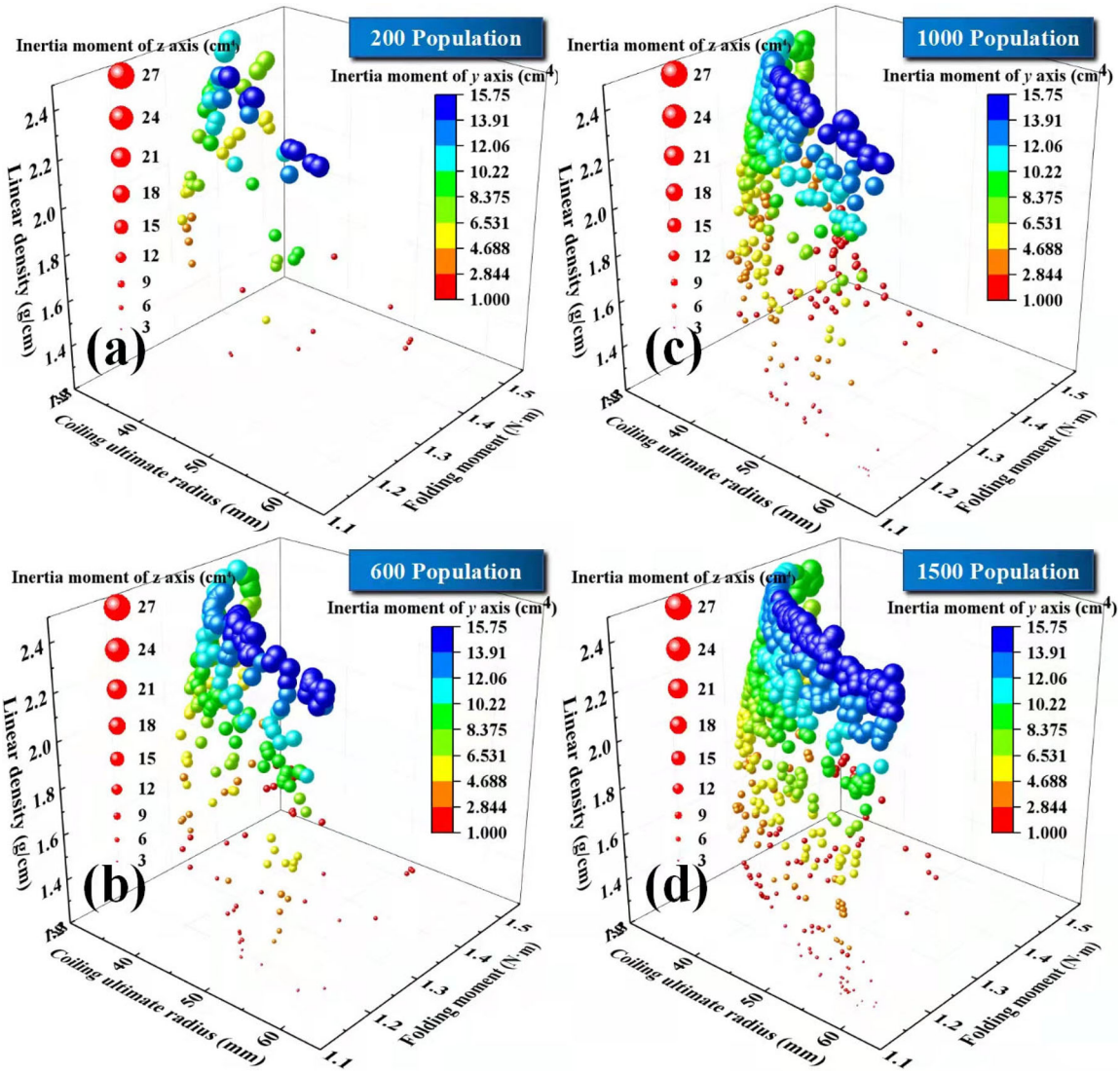


Figure 6. Comparison of Pareto fronts for different populations sizes: (a) 200 Population (b) 400 Population (c) 1000 Population (d) 1500 Population.

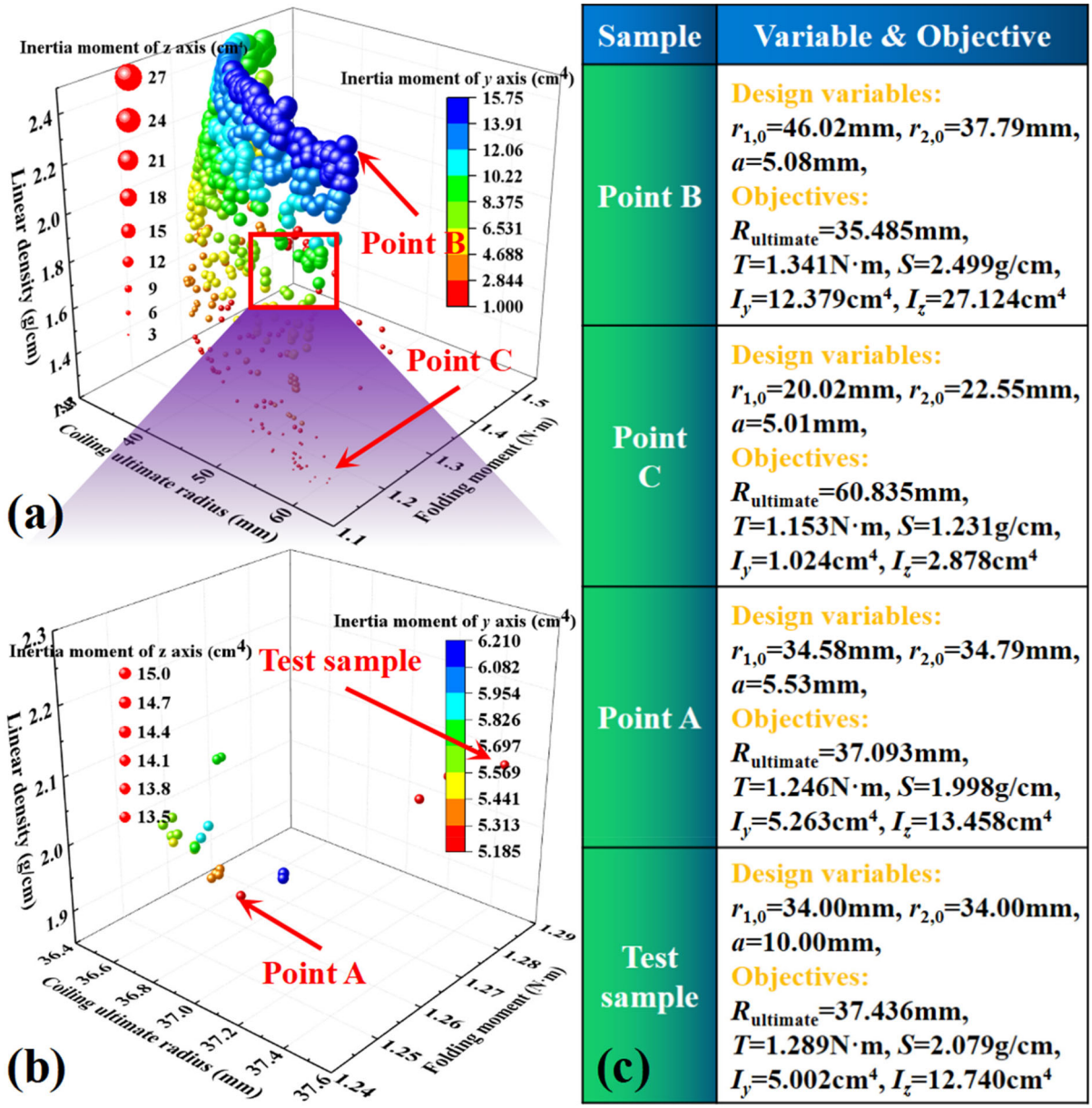


Figure 7. Optimal design points of the LDC boom: (a) 1365 design points (b) 48 design points and test sample (c) Point A, Point B, Point C and test sample.

The optimization results show that a total of 48 design points are found on the Pareto front, all of which are better than test sample for the five objectives (i.e. the minimum ultimate coiling radius, the minimum folding moment, the minimum linear density and the maximum inertia moments of y axis and z axis of the cross-section). These 48 design points are listed in Appendix A and Figure 7b. To further illustrate the optimization results, a design point was selected by hand from the Pareto front, namely Point A. The comparison results of test sample and Point A are shown in Figure 7c. According to Figure 7c, compared with test sample (R_{ultimate}=37.436 mm, T = 1.289N·m, S = 2.079g/cm, I_y=5.002cm^4 and I_z=12.740cm^4), Point A (R_{ultimate}=37.093 mm, T = 1.246N·m, S = 1.998g/cm, I_y=5.263cm^4, I_z=13.458cm^4) is a better design scheme. The ultimate coiling radius is reduced by 0.92%, the folding moment decreases by 3.33%, the linear density reduces 3.90%, while the inertia moments of y axis and z axis of the cross-section increase by 5.22% and 5.64%, respectively.

In order to further illustrate the effectiveness of the multi-objective optimization design framework proposed in this paper, two points with special positions on the Pareto front were selected by hand, namely Point B and Point C. The comparison results of test sample, Points B and C are shown in Figure 7c. Compared with test sample, the inertia moments of y axis and z axis of the cross-section of Point B increase significantly, of which the inertia moment of y axis of the cross-section ascends by 147.48% and the inertia moment of z axis of the cross-section ascends by 112.90%. Compared with test sample, the linear density of Point C is greatly dropped by 40.79%. Therefore, the many-objective optimization design framework proposed in this paper can effectively optimize the LDC boom.

Table 6. 22 design points on the Pareto front.

Design No.	$r_{1,0}$(mm)	$r_{2,0}$(mm)	a(mm)	$R_{ultimate}$ (mm)	T(N·m)	S(g/cm)	I_y(cm^4)	I_z(cm^4)
1	20.02	22.55	5.01	60.835	1.153	1.231	1.024	2.878
2	24.73	28.29	5.01	47.164	1.164	1.496	1.933	5.502
3	23.93	39.64	5.33	48.729	1.259	1.597	1.781	7.145
4	33.35	20.63	5.59	58.183	1.118	1.772	4.679	8.513
5	35.77	20.51	5.20	58.687	1.120	1.859	5.774	10.053
6	38.35	20.88	5.26	57.253	1.134	1.970	7.109	12.033
7	35.42	35.03	5.20	36.840	1.248	2.028	5.654	14.261
8	38.10	29.75	5.32	40.630	1.213	2.076	7.003	14.799
9	40.94	25.33	5.08	46.123	1.184	2.130	8.661	15.754
10	41.88	25.57	5.76	45.741	1.201	2.189	9.271	16.731
11	44.53	21.63	5.01	54.650	1.171	2.225	11.124	17.762
12	37.61	47.58	5.19	35.558	1.383	2.266	6.823	21.215
13	43.24	30.37	5.33	40.075	1.250	2.295	10.219	20.107
14	45.37	25.22	5.23	46.309	1.213	2.312	11.776	20.157
15	46.91	22.98	5.63	50.857	1.207	2.355	13.003	20.852
16	47.45	25.04	5.01	46.621	1.223	2.389	13.466	22.417
17	44.08	37.30	5.04	35.695	1.321	2.410	10.861	24.307
18	41.97	46.90	5.07	33.892	1.404	2.439	9.440	26.332
19	49.07	25.23	5.23	46.303	1.238	2.463	14.888	24.454
20	44.43	41.56	5.13	34.029	1.368	2.479	11.143	26.895
21	48.76	28.43	5.05	41.966	1.266	2.488	14.618	25.691
22	48.08	30.94	5.32	39.589	1.289	2.500	14.033	26.136

Table 7. Parameter definition for four other state-of-the-art algorithms.

Algorithms	NSGA-II	MOEA/D	awGA	RVEA
No. of populations	1500	1500	1500	1500
No. of generations	200	N/A	200	200
Stop criterion	Reach the max. generation (200)	Reach the total number of fitness evaluation	Reach the max. generation (200)	Reach the max. generation (200)
Crossover rate	0.7	0.7	0.7	0.7
Mutation rate	0.08	0.08	0.08	0.8
Selector operator	NSGA-crowding distance	Steady-state selection scheme	Adaptive adjustment of weights	Angle-based selection criterion
Elitism operator	Active	Active	Active	Active

7. Determining the best benchmarked algorithm

The 'no free lunch' theorem illustrates that an algorithm that improves its performance on a category of problems inevitably degrades its performance on other problems. Although the many-objective optimization design framework presented in this paper selects a well-known general algorithm, NSGA-III, the general algorithm may not show the good performance for specific problems. Bai and his coworkers [2, 4] have proved that multi-objective evolutionary algorithm based on decomposition (MOEA/D) based on decomposition shows better performance in some composite optimization problems than NSGA-II and its development version based on non-dominated sorting. However, MOEA/D shows the worst performance in optimizing triaxial weave fabric composites under tension [25]. Therefore, it is necessary to benchmark other state-of-the-art algorithms.

7.1. Five state-of-the-art algorithms

For the multi-objective optimization problem in this research, four other state-of-the-art many-objective algorithms in Geatpy toolbox were used to optimize the LDC boom, including NSGA-II, MOEA/D, adaptive-weight genetic algorithm (awGA) and reference vector guided evolutionary algorithm (RVEA). The hyperparameters of four other state-of-the-art algorithms utilized in this optimization problem are listed in Table 7, in which the selection, crossover and mutation types are default settings. In order to perform a fair benchmarking across the five algorithms, the same genetic operator types of selection, crossover and mutation were used, as well as the same operator rate which is selected for the CEC'09 benchmarking. Each algorithm is run over 30 independent cycles to obtain a statistically significant mean value due to the stochastic nature of these algorithms.

Figure 7 and Figure 8a–8d show the optimization results of the Pareto front of the five algorithms. NSGA-III, NSGA-II, MOEA/D and awGA all find Pareto fronts with similar coverage, quality and diversity. It is further illustrated that the many-objective optimization design framework established in this research is effective for optimizing the LDC boom. However, RVEA is expected to have excellent performance as a general algorithm but the performance is poor, and only dozens of design points are found on the Pareto front.

7.2. Performance metrics

To quantitatively evaluate the performance, two metrics are introduced in the work which are the inverted generational distance (mIGD) and hyper volume (mHV). The mIGD is determined by using Eq. (66) as [25],

$$mIGD(O, M^*) = \frac{\sum_{\nu \in M} d(\nu, O)}{|M^*|}, \tag{66}$$

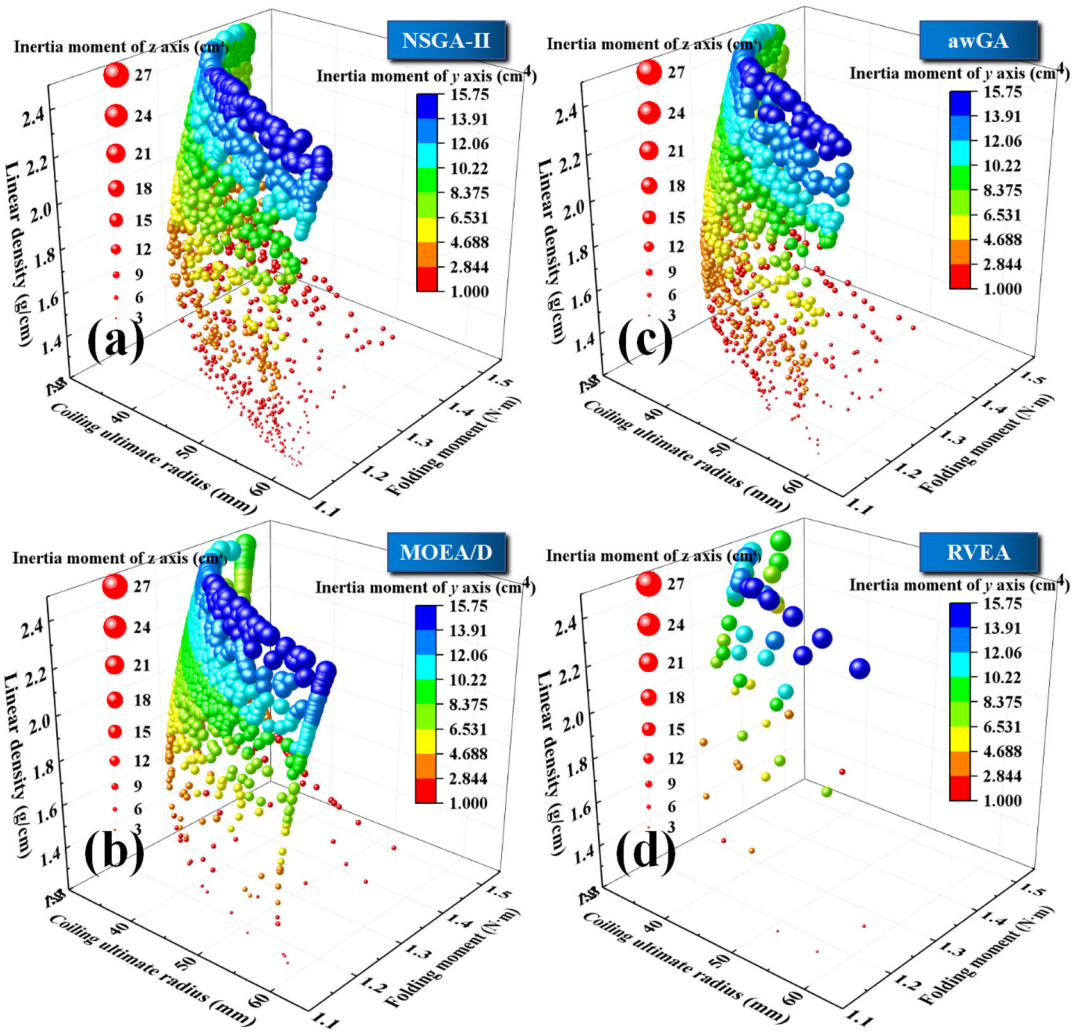


Figure 8. Comparison of Pareto fronts for different state-of-the-art algorithms: (a) NSGA-II (b) MOEA/D (c) awGA (d) RVEA.

where M^* represents a set of points along the mimicked Pareto front, O represents a set of points from the currently obtained Pareto front, ν represents each point in the set M^* and $d(\nu, O)$ calculates the minimum Euclidean distance between ν and the points in O. The lower mIGD value reflects the better convergence, accuracy and diversity of the Pareto front.

The mimicked Pareto front is created using the non-dominated solutions from all of the Pareto fronts from the 150 simulations. The mHV [26] is inspired by the HV indicator. The HV indicator calculates the summation of the objective space volume between the obtained Pareto front points and a predefined reference point. The predefined reference point in the mHV is developed in the same manner as the mimicked Pareto Front for mIGD. A higher mHV value indicates an algorithm with a better performance. When comparing the results from the two metrics, the mHV metric is more representative of the diversity of the population but mIGD is more representative of the accuracy and uniformity of the obtained Pareto fronts, which is especially true when there are a small number of points in the Pareto front. Therefore, using both metrics together allows an evaluation of the accuracy and the diversity of the obtained Pareto fronts for each algorithm.

To evaluate the convergence of each of the benchmarked algorithms, the Pareto front results were extracted from each run at every 75,000 function calls, labeled as 50 generations, until 300,000 function calls, labeled as 200 generations. The convergence of the mIGD and mHV values for the five algorithms are illustrated in Figure 9. NSGA-II achieves the best performance on the five-objective optimization problem, with a mean mIGD of 0.0249 and a mean mHV of 0.9160 at 200 generations, and NSGA-III ranks the second place with both metrics. Similarly, the top two algorithms both converge at 150 generations. awGA does not perform well at 50 generations, mIGD values decreases and mHV values increases with the increased generation number. awGA achieves the third and it is expected to perform better when generation is greater than 200. Although MOEA/D can realize rapid convergence, it shows high mean mIGD and low mHV, so MOEA/D ranks fourth. This provides a contradiction to the evolutionary computation literature where MOEA/D has previously been illustrated to have a dominant performance on many-objective problems [2, 4]. It is clearly

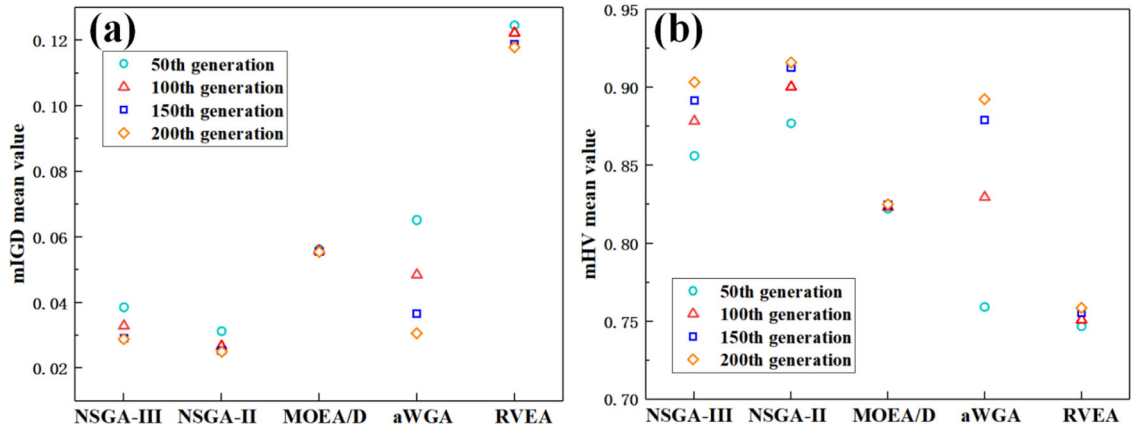


Figure 9. Comparison of mIGD and mHV mean values for five state-of-the-art algorithms: (a) mIGD mean value (b) mHV mean value.

Table 8. Friedman test ranks for five state-of-the-art algorithms.

	mIGD					mHV				
Algorithms	Min.	Max.	Mean.	SD	Rank	Min.	Max.	Mean.	SD	Rank
NSGA-III	0.0265	0.0303	0.0288	0.0013	2	0.8938	0.9005	0.8961	0.0018	2
NSGA-II	0.0221	0.0271	0.0249	0.0011	1	0.9138	0.9188	0.9160	0.0014	1
MOEA/D	0.0505	0.0598	0.0555	0.0026	4	0.8201	0.8301	0.8251	0.0025	4
awGA	0.0288	0.0336	0.0307	0.0012	3	0.8887	0.8955	0.8924	0.0016	3
RVEA	0.1085	0.1246	0.1179	0.0049	5	0.7545	0.7632	0.7588	0.0020	5

Table 9. Comparison of mIGD and mHV values between any two algorithms.

	mIGD			mHV		
Comparison of algorithms	*p*-value	|*d*|	|*r*|	*p*-value	|*d*|	|*r*|
NSGA-III vs NSGA-II	0.014	3.238	0.850	0.004	12.341	0.987
NSGA-III vs MOEA/D	<0.001	12.989	0.988	<0.001	32.594	0.998
NSGA-III vs awGA	0.014	1.518	0.604	0.102	2.172	0.735
NSGA-III vs RVEA	<0.001	24.855	0.996	<0.001	72.163	0.999
NSGA-II vs MOEA/D	<0.001	15.328	0.991	<0.001	44.864	0.999
NSGA-II vs awGA	<0.001	5.038	0.929	<0.001	15.698	0.991
NSGA-II vs RVEA	<0.001	26.189	0.997	<0.001	91.063	0.999
MOEA/D vs awGA	0.014	12.247	0.986	0.004	32.065	0.998
MOEA/D vs RVEA	0.014	17.803	0.994	0.014	29.286	0.997
awGA vs RVEA	<0.001	15.908	0.992	<0.001	73.768	0.999

|*d*|: absolute value of effect size.
|*r*|: absolute value of Pearson correlation coefficient.

from Figure 9a and 9b that RVEA exhibits the worst performance among the five benchmarked algorithms.

7.3. Statistical analyses

The 30 results of five algorithms with 1500 populations and 200 generations were selected to further quantify the performance. Joint analysis of results achieved by various algorithms was estimated by Friedman test, which is a nonparametric test and equivalent to the analysis of variance with repetitive sizes (within groups). It performs multiple comparisons to predict significant difference between two or more algorithms. This test was conducted using SPSS software (SPSS Statistics, Version 26, IBM Corporation, New York). Friedman test ranks for five state-of-the-art algorithms are listed in Table 8. In terms of standard deviation (SD), all algorithms have small values, illustrating that the dispersion of the date is low. From Friedman test ranks performed for all algorithms, NSGA-II achieves the best performance, whereas the NSGA-III, awGA, MOEA/D and RVEA settle at second, third, fourth and fifth rank, respectively, which is consistent with the ranking in Section 7.2.

To further quantify the degree of difference between two algorithms, the effect size (ES) was calculated using Cohen's d [27]. Values between 0.21 and 0.50 denoted a small effect size, those between 0.51 and 0.80 denoted a moderate effect size, and those >0.80 denoted a large effect size. In addition, the Pearson correlation was utilized to examine the correlation between two algorithms. A *p*-value of <0.05 was interpreted as indicating a statistically significant change. The comparison of mIGD and mHV values between any two algorithms are listed in Table 9. The results report that most algorithms have significantly different distributions of Pareto front results when solving the five optimization problems. Only mHV values of NSGA-III and awGA do not show significantly different dispersion.

8. Results and discussion

8.1. Analytical models

The folding moment and the ultimate coiling radius are the key factors to evaluate the folding behavior of the LDC boom. Generally, researchers use experimental method and numerical simulation to study the two mechanical properties. However, the cost of measuring the two mechanical properties by the experimental method is relatively high, and the experimental process is easy to be affected by many accidental factors. Moreover, as a general tool widely used to solve various engineering problems, numerical simulation usually has high prediction accuracy. But numerical simulation needs to establish a complex finite element model, which bring challenges to designers. Analytical models proposed in Sections 3 and 4 accurately predicts the folding moment and the ultimate coiling radius the LDC boom. These analytical models are practical and convenient, and only few geometric parameters and mechanical properties of constituent materials are required to predict the folding moment the LDC boom, which has a good practical value in engineering.

8.2. Optimization design

The many-objective optimization design framework proposed in this paper can effectively optimize the LDC boom and help designers select different design schemes according to practical engineering requirements. As shown in Section 6, there are a total of 1365 design points on the entire Pareto front. Different design schemes have different space applications. On the Pareto front, there are a total of 48 design solutions, all of which are better than test sample in the literature [15] for the five objectives. As a basic part of space deployable structures, the LDC boom usually should have high stiffness, high flexural stiffness and high natural frequency in space applications. Therefore, it is suggested that designers pay more attention to those design schemes with large inertia moments of y axis and z axis of the cross-section, such as Point B. In addition, NASA has also recently expressed an interest in the light and small LDC boom as a candidate solar sail boom for low-cost deep space exploration [11], thus allowing designers to focus on those design points with small occupied space volume and low linear density, such as Point C. Furthermore, five state-of-the-art algorithms were benchmarked through two performance metrics and Friedman test, which shown that NSGA-II and NSGA-III ranked the top two among the five algorithms, respectively. Using the proposed guideline, the efficiency of searching the optimal designs will be significantly increased in the practice.

9. Conclusions

Due to low mass-to-volume ratio, high packaging efficiency and excellent mechanical properties, the LDC boom is being increasingly applied in space field. This paper aimed to establish analytical models for predicting the folding behavior of the LDC boom and propose a many-objective optimization design framework. Three important results emerging from the research are as follows

1. An analytical model for predicting the folding moment of the LDC boom in the folding process was established. The energy principle was utilized to determine the folding moment of the LDC boom in the folding process. The folding moment predicted by the analytical model agrees well with the experimental result in the literature, indicating that the analytical model shows good prediction accuracy.
2. An analytical model for predicting the ultimate coiling radius of the LDC boom in the folding process was established according to the classical laminate theory and the maximum stress criterion. The predicted ultimate coiling radius of a LDC boom using the analytical model is smaller than the practical coiling radius of the LDC boom without failure, which is consistent with the experiment. The analytical model for predicting the ultimate coiling radius of the LDC boom is effective and feasible.
3. Based on the classical NSGA-III algorithm, a many-objective optimization design framework was proposed for optimizing the LDC boom. The optimization results shows that a total of 48 design points are found on the Pareto front, all of which are better than test sample for the five objectives. It is helpful for designers to determine required design solutions. The many-objective optimization design framework proposed in this paper can effectively optimize the LDC boom. However, it is clear that optimization results of the 48 design points have not achieved significant improvement compared with the test sample. The reason behind this phenomenon is that the many-objective design framework presented in this paper sets many objectives, and each objective needs to be weighed in the optimization process. Compared with single-objective and bi-objective optimization, the complexity of many-objective optimization is significantly increased. Through insight into the current optimization results, the optimization results are expected to perform better if some secondary objectives are removed.
4. Five state-of-the-art algorithms were benchmarked. NSGA-II and NSGA-III ranked the top two among the five algorithms, which was determined by quantitative analysis methods, including mIGD, mHV and Friedman test. Optimizing the LDC boom using NSGA-II is expected to obtain better optimization results. As a general algorithm, RVEA was expected to provide top performance, but RVEA exhibited poor performance. Without investigating the dominant characteristic, the selection of the algorithm is critical to find the optimal solutions.

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Appendix A. 48 optimal design points

Design No.	$r_{1,0}$(mm)	$r_{2,0}$(mm)	***a*(mm)**	$R_{ultimate}$(mm)	T(N·m)	S(g/cm)	I_y(cm4)	I_z(cm^4)
1	34.58	34.79	5.53	37.093	1.246	1.998	5.263	13.458
2	34.58	34.79	5.53	37.093	1.246	1.998	5.263	13.458
3	34.81	34.90	5.38	36.960	1.246	2.005	5.370	13.695
4	34.81	34.90	5.38	36.960	1.246	2.005	5.370	13.695
5	34.82	34.84	5.49	36.959	1.247	2.008	5.371	13.677
6	34.82	34.79	5.53	36.972	1.247	2.008	5.373	13.665
7	34.82	34.79	5.53	36.972	1.247	2.008	5.373	13.665
8	34.82	34.79	5.53	36.972	1.247	2.008	5.373	13.665
9	34.82	34.91	5.53	36.956	1.248	2.010	5.374	13.703
10	34.82	34.91	5.53	36.956	1.248	2.010	5.374	13.703
11	35.22	35.22	5.16	36.736	1.248	2.020	5.559	14.147
12	35.22	35.22	5.16	36.736	1.248	2.020	5.559	14.147
13	35.22	35.22	5.16	36.736	1.248	2.020	5.559	14.147
14	35.22	35.22	5.16	36.736	1.248	2.020	5.559	14.147
15	35.22	35.22	5.16	36.736	1.248	2.020	5.559	14.147
16	35.22	35.22	5.16	36.736	1.248	2.020	5.559	14.147
17	35.22	35.22	5.16	36.736	1.248	2.020	5.559	14.148
18	35.28	35.46	5.00	36.705	1.249	2.022	5.588	14.279
19	35.31	35.03	5.22	36.838	1.248	2.023	5.599	14.160
20	35.31	35.03	5.22	36.838	1.248	2.023	5.599	14.160
21	35.31	35.03	5.22	36.838	1.248	2.023	5.599	14.160
22	35.31	35.03	5.22	36.838	1.248	2.023	5.599	14.160
23	35.39	35.46	5.00	36.641	1.250	2.027	5.643	14.382
24	35.39	35.46	5.00	36.641	1.250	2.027	5.643	14.382
25	35.42	35.03	5.20	36.840	1.248	2.028	5.654	14.261
26	35.42	35.03	5.20	36.840	1.248	2.028	5.654	14.261
27	35.42	35.03	5.20	36.840	1.248	2.028	5.654	14.261
28	35.42	35.03	5.20	36.840	1.248	2.028	5.654	14.261
29	35.42	35.03	5.20	36.840	1.248	2.028	5.655	14.261
30	35.42	35.03	5.20	36.839	1.248	2.028	5.655	14.262
31	35.39	35.22	5.16	36.734	1.249	2.028	5.642	14.301
32	34.37	39.12	5.09	37.214	1.280	2.030	5.189	14.703
33	34.37	39.12	5.09	37.214	1.280	2.030	5.189	14.703
34	35.39	35.46	5.16	36.641	1.252	2.031	5.643	14.382
35	35.39	35.46	5.17	36.641	1.252	2.031	5.643	14.382
36	35.69	34.99	5.00	36.864	1.247	2.033	5.781	14.479
37	35.69	34.99	5.00	36.864	1.247	2.033	5.781	14.479
38	34.38	40.09	5.01	37.211	1.288	2.039	5.195	15.032
39	35.92	34.97	5.04	36.871	1.248	2.044	5.893	14.680
40	35.89	35.03	5.20	36.840	1.251	2.047	5.880	14.675
41	35.59	36.68	5.33	36.535	1.266	2.058	5.744	14.977
42	35.59	36.73	5.33	36.535	1.266	2.059	5.744	14.995
43	35.59	36.80	5.33	36.535	1.267	2.059	5.744	15.018
44	36.53	34.20	5.10	37.319	1.244	2.061	6.191	14.954
45	36.53	34.22	5.10	37.301	1.245	2.061	6.191	14.964
46	36.53	34.23	5.10	37.300	1.245	2.061	6.191	14.965
47	36.55	34.22	5.10	37.301	1.245	2.062	6.206	14.990
48	36.55	34.22	5.10	37.301	1.245	2.062	6.206	14.990